Waves

Building Models to Describe our World
Wave Properties, Descriptions and Phenomena

Outline

Properties of a Wave

Building Models to Describe our World

Wave Properties, Descriptions and Phenomena

Speed of a wave

- Depends only on the medium through which it propagates (e.g. speed of sound does not depend on what is making the sound).
- If the particles in the medium are heavier, they have more inertia and they oscillate about their equilibrium slower $\rightarrow$ Waves are slower in denser media.
- If the medium is "stiffer" (the force trying to restore the particles to equilibrium is larger), they will oscillate faster $\rightarrow$ Waves are faster in stiffer media.

- For waves on a rope with tension and linear mass density:

$$v = \sqrt{\frac{F_T}{\mu}}$$

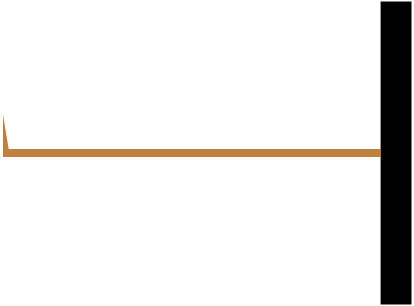
Analogous to a simple harmonic oscillator

$$\omega = \sqrt{\frac{k}{m}} = 2\pi f$$

- For a stiffer spring constant, the oscillator has a higher frequency.
- For a larger mass, the oscillator has a lower frequency.

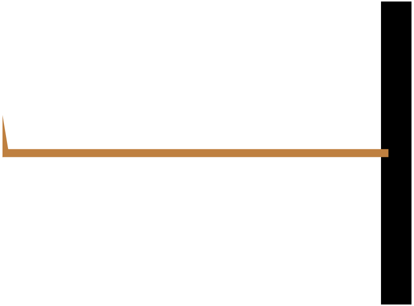
Pulse reflection

Fixed end, reflection inverted:



Think of rope exerting upwards force on wall, so wall exerts downwards force on rope, which starts a negative pulse going to the left

Open end, reflection in the same direction



The end of the rope moves up when pulse arrives, which starts a positive pulse going to the left

Reflection and transmission

- When a pulse changes medium, part of it is reflected and part is transmitted.

When the second section is heavier, it reflect the same as a fixed end.

- The amplitude (and sign) of the reflected part is related to the reflection coefficient:

$$R = \frac{\sqrt{\mu}_1 - \sqrt{\mu}_2}{\sqrt{\mu}_1 + \sqrt{\mu}_2}$$

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Describing a Wave

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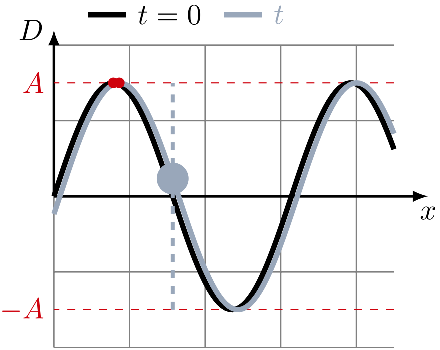
How to describe a wave

- A wave is a series of pulses. The easiest type of wave to describe is one where the pulses are generated by a periodic mechanism (e.g. the end of the rope moving up and down at constant frequency).
- If the end of the rope moves like a simple harmonic oscillator, then you obtain a sine wave.

→ To fully describe a wave, we need to describe the position of each point in the medium as a function of time.

→ $D(x,t)$, distance of the point at position x from its equilibrium position at time t.

→ $D(x,t)$ depends on **position and time**.



A sine wave propagating at constant speed through a medium. Each point in the medium executes SHM.

At time $t=0$, we have:

$$D(x,t=0) = A\sin\left(\frac{2\pi}{\lambda}x\right)$$

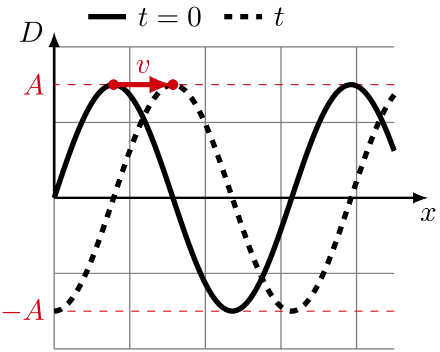
Traveling wave description

- At time $t = 0$, we can describe the displacement of each point in the medium by:

$$D(x, t = 0) = A \sin\left(\frac{2\pi}{\lambda} x\right)$$

- At time t, the point at position x will have the same displacement as the point at $x - vt$ had at $t = 0$:

$$\begin{aligned} D(x, t) &= D(x - vt, t = 0) \\ &= A \sin\left(\frac{2\pi}{\lambda}(x - vt)\right) \end{aligned}$$



When a sine wave propagates through the medium, each point in the medium moves back and forth about some equilibrium position.

Description of a wave using wavelength and period

$$D(x,t) = A\sin\left(\frac{2\pi}{\lambda}(x - vt)\right)$$

- The speed of the wave, v, is related to its wavelength, λ, and period, T:

$$v = \frac{\lambda}{T}$$

$$\therefore D(x,t) = A\sin\left(\frac{2\pi}{\lambda}x - \frac{2\pi}{T}t\right)$$

General description, phase, wave number

- In general, the displacement need not be $D = 0$ at $x = 0$ and time $t = 0$. The wave can be "shifted" to the left or to the right by a "phase", ϕ:

$$D(x,t) = A\sin\left(\frac{2\pi}{\lambda}x - \frac{2\pi}{T}t + \phi\right)$$

- Clean up the notation to get rid of the 2π:

$$D(x,t) = A\sin\left(kx - \omega t + \phi\right)$$

  by introducing wave number, k, and angular frequency, ω:

$$k = \frac{2\pi}{\lambda}$$
$$\omega = \frac{2\pi}{T} = 2\pi f$$

Wave number and angular frequency are directly related to wavelength and period/frequency, simply a cleaner version without the 2π.

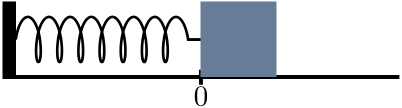
Recall: simple harmonic motion

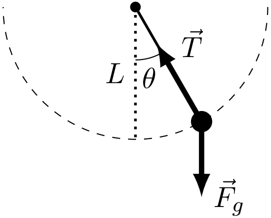
- Any system described by the following undergoes Simple Harmonic Motion:

$$\frac{d^2x}{dt^2} = -\omega^2 x$$
$$\therefore x(t) = A\cos(\omega t + \phi)$$

- SHM arises from physics where there is a *restoring* force that is *linearly* proportional to displacement. For example, $F = -kx$.
- All of the physics is contained in ω, the angular frequency of the motion:

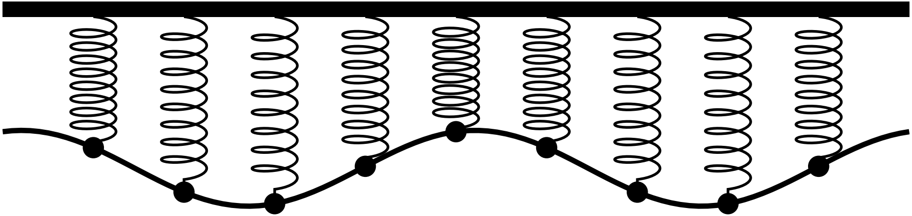
$$\omega = \sqrt{\frac{k}{m}} \quad , \quad \omega = \sqrt{\frac{g}{L}}$$





A spring-mass system or a simple pendulum are examples of systems that undergo simple harmonic motion.

A sine wave and SHM



A sine wave going through a medium corresponds to particles in the medium being displaced in SHM.

The wave equation (1D)

- Similarly, any physics model that gives "the wave equation":

$$\frac{\partial^2 D}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 D}{\partial t^2}$$

  corresponds to the propagation of a wave.
- A possible mathematical description/solution is:

$$D(x,t) = A\sin\left(kx - \omega t + \phi\right)$$

  Where:
  - A is the amplitude of the wave and depends on what is creating the wave (e.g. how high you shake the end of a rope).
  - ω and k (or equivalently T and λ) are related by:

$$\frac{\omega}{k} = \frac{\lambda}{T} = v$$

  Only one of ω and k is independent, since the speed, v, depends on the medium and not what creates the wave.

Wave equation solutions, linear superposition

- An infinite number of functions, $D(x,t)$, can solve the wave equation:

$$\frac{\partial^2 D}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 D}{\partial t^2}$$

  For example:

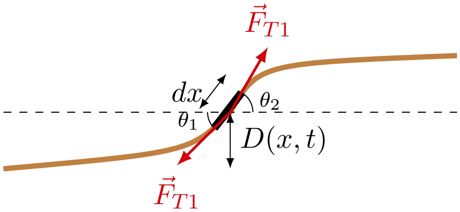
$$\begin{aligned} D_1(x,t) &= A_1 \sin\left(k_1 x - \omega_1 t + \phi_1\right) \\ D_2(x,t) &= A_1 \cos\left(k_2 x - \omega_2 t + \phi_2\right) \\ &\ldots \end{aligned}$$

- You can also show that any linear combination of these functions (or other possible solutions) will also satisfy the wave equation:

$$D_3(x,t) = a_1 D_1(x,t) + a_2 D_2(x,t)$$

A physics model for a pulse on a rope with tension

We can model the vertical acceleration of a small piece of rope of length dx:



Modeling a pulse on a rope.

A physics model for a pulse on a rope with tension

We can model the vertical acceleration of a small piece of rope of length dx:

- This leads to the wave equation in the form:

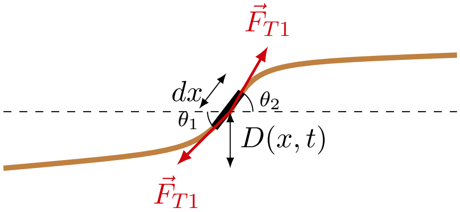
$$\frac{\partial^2 D}{\partial x^2} = \frac{\mu}{F_T}\frac{\partial^2 D}{\partial t^2}$$

- Compare with the wave equation:

$$\frac{\partial^2 D}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 D}{\partial t^2}$$

and identify that:

$$v = \sqrt{\frac{F_T}{\mu}}$$



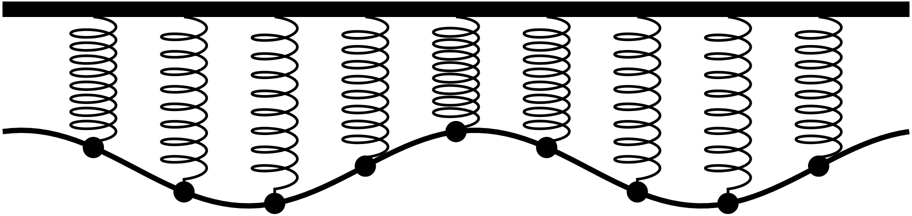
Modeling a pulse on a rope.

Energy Transport and Phenomena of Waves

Building Models to Describe our World

Wave Properties, Descriptions and Phenomena

Energy transport in a wave



A sine wave going propagating through a medium modeled as a collection of SHOs.

We can think of each point in the medium as a simple harmonic oscillator that moves up and down with the **same period/frequency and the same amplitude** as the wave traveling through the medium.

Energy transport in a 1D wave

- Energy of one oscillator:

$$E = \frac{1}{2}\omega^2 m A^2$$

- If the oscillator is a small mass element on a rope:

$$dm = \mu dx$$

with energy:

$$dE = \frac{1}{2}\omega^2 dm A^2 = \frac{1}{2}\omega^2 \mu dx A^2$$

- The energy "contained" in 1 wavelength is the sum of the energies of the oscillators in one wavelength:

$$E_\lambda = \int dE = \int_0^\lambda \frac{1}{2}\omega^2 \mu dx A^2 = \frac{1}{2}\omega^2 \mu A^2 \lambda$$

Power transport in a 1D wave

- Power is the rate at which energy is being expended:

$$P = \frac{\Delta E}{\Delta t}$$

- Over one period of the wave, T, you need to expend the energy "contained" in one wavelength:

$$P = \frac{\frac{1}{2}\omega^2 \mu A^2 \lambda}{T} = \frac{1}{2}\omega^2 \mu A^2 v$$

  since:

$$v = \frac{\lambda}{T}$$

- This is the power that you have to put in to sustain the wave (and the rate at which energy is "received" at the other end).

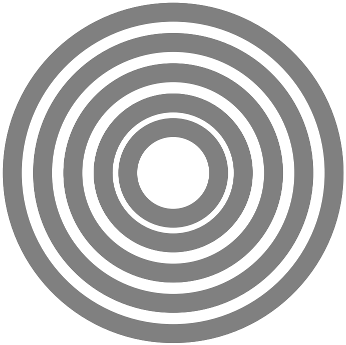
Energy transport in a 3D spherical wave

- A spherical sound wave corresponds to successive shells of air contracting and expanding (longitudinal wave).
- We can model each shell as a simple harmonic oscillator with mass dm and spring constant k_s.
- If each shell has the same thickness, dr, then the outer shells have higher mass:

$$dm = \rho dV = \rho 4\pi r^2 dr$$

- The energy in a shell is then given by:

$$dE = \frac{1}{2}k_s A^2 = \frac{1}{2}\omega^2 dm A^2 = 2\pi\rho\omega^2 A^2 r^2 dr$$



A spherical longitudinal wave propagating through successive concentric shells, each with thickness, dr.

Energy transport in a 3D spherical wave

- Energy in one shell:

$$dE = 2\pi\rho\omega^2 A^2 r^2 dr$$

- Power (rate at which energy moves from one shell to the next):

$$P = \frac{dE}{dt} = \frac{dE}{dr}\frac{dr}{dt} = \frac{dE}{dr}v = 2\pi\rho\omega^2 A^2 r^2 v$$

Intensity of a wave

- The intensity of a spherical wave is the power that it delivers into a specific area (for example, the area of your ear for a sound wave).
- At some distance r from the source, the total power per unit area is given by:

$$I = \frac{\text{Power}}{\text{Area}} = \frac{P}{4\pi r^2} = \frac{1}{2}\rho\omega^2 A^2 v \propto \frac{1}{r^2}$$

- Your ear is sensitive to the intensity of a sound wave (as your house is sensitive to the intensity of an earthquake).
- It is sensitive to the rate at which energy is delivered to your ear drum.
- Since the amplitude goes down as $1/r$, the intensity of the wave goes down as $/r^2$.

Superposition of waves

Recall that many functions can solve the wave equation:

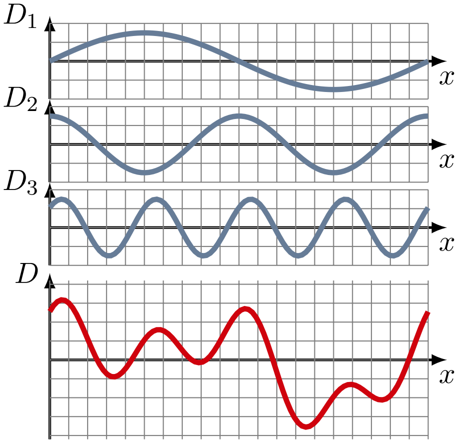
$$\frac{\partial^2 D}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 D}{\partial t^2}$$

In particular, if $D_1(x,t)$ and $D_2(x,t)$ are solutions, so is any linear combination thereof:

$$D_3(x,t) = a_1 D_1(x,t) + a_2 D_2(x,t)$$

Superposition and Fourier's theorem

- If you have several mechanisms generating waves, the resultant waves is the sum of each individual wave.
- **Fourier's theorem**: Any periodic function can always be expressed as a sum of sinusoidal functions. $\rightarrow$ Thus, we can study sinusoidal waves without any loss in generality.



The bottom wave, D, is the superposition of the 3 sinusoidal waves, $D = D_1 + D_2 + D_3$.

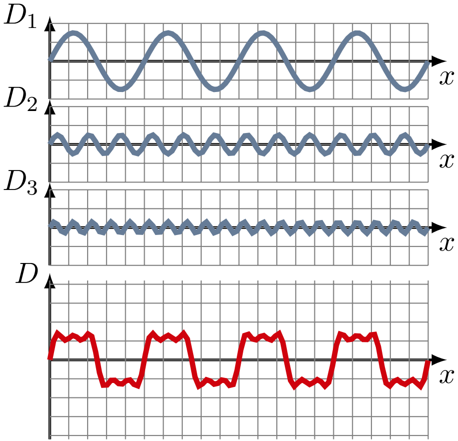
Example: Fourrier decomposition of a square wave

- Even a square wave can be modeled as the sum of sine waves:

$$f(x) = \frac{4}{\pi} \sum_{n=1,3,5,\ldots}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi x}{L}\right)$$

  where the square wave has wavelength $2L$ and amplitude of 1.

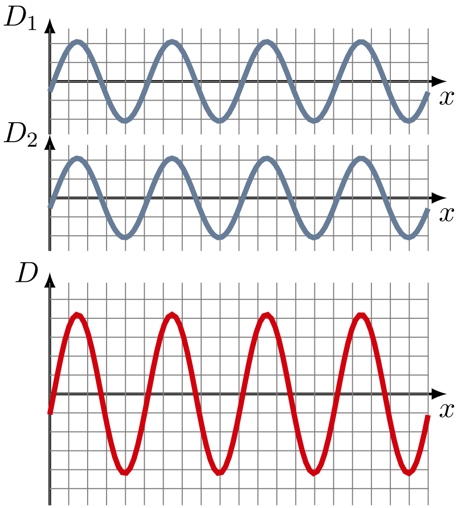
$\rightarrow$ Without loss of generality, we can worry only about sine waves!



Adding sine waves to make a square wave. This converges in the limit of an infinite sum.

Interference

- Sometimes, the waves that sum together “interfere” in “constructive” or “destructive” ways.
- To get a complete interference, you need waves of the same frequency/wavelength to interfere.



Constructive interference between two identical waves in phase.

Interference

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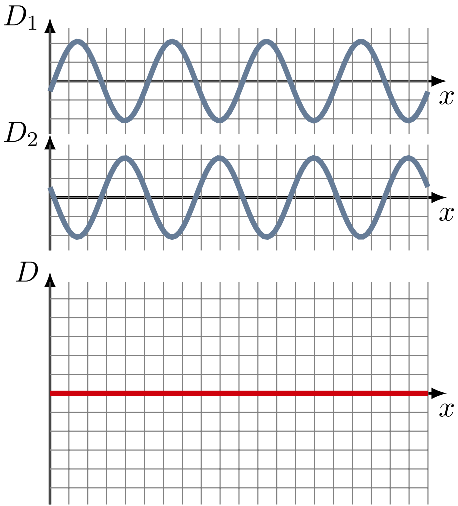
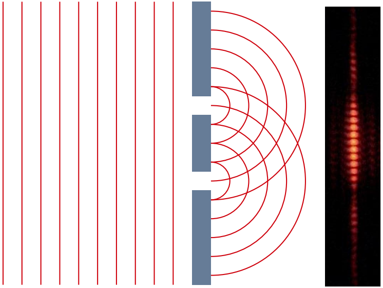
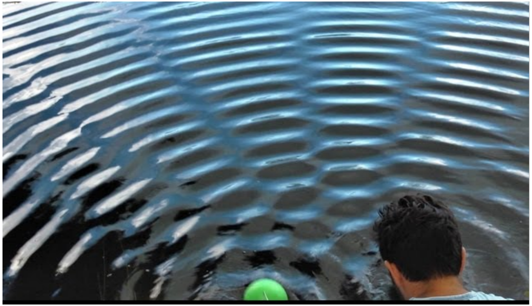


Destructive interference between two identical waves out of phase.

Interference, double-slit



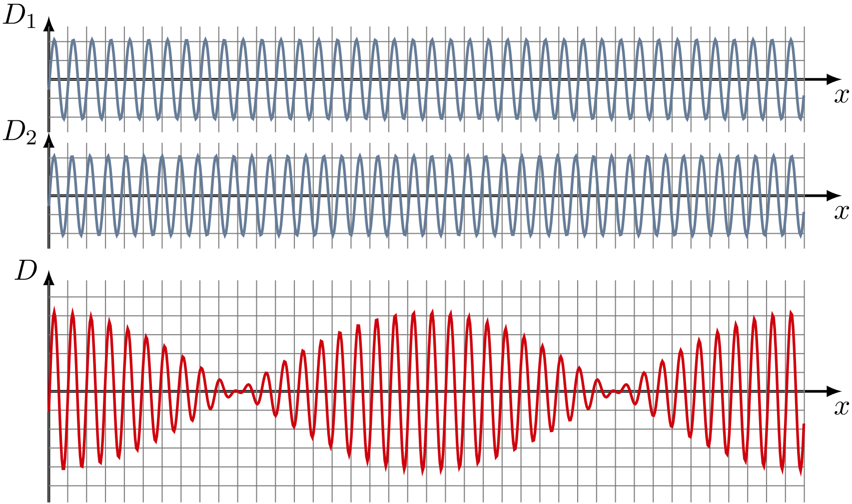
The interference pattern from light shining through two slits (double-slit experiment).



Veritasium (former Queen's student!). Waves are created by two ball oscillating on top of the water.
https://www.youtube.com/watch?v=Iuv6hY6zsd0

Waves from two sources

Speakers interfering and beats

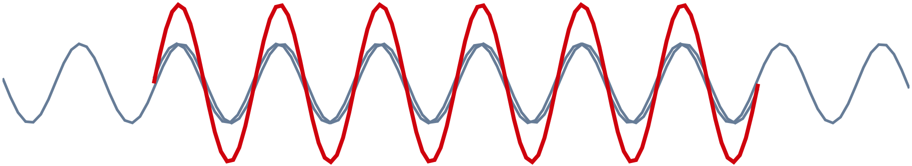


Waves with slightly different frequencies result in “beats” (here the frequencies are different by 5%).

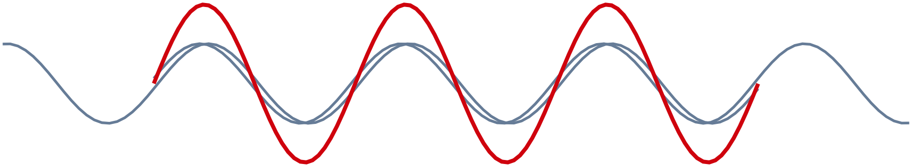
Standing waves

- Particles in the medium are oscillating, but energy is not moving through the medium (as opposed to traveling waves).
- This happens when waves traveling in opposite directions interfere, usually with their own reflections.
- Examples:
  - Vibrating guitar/piano/violin string.
  - Vibrating air inside an organ pipe.

Standing waves



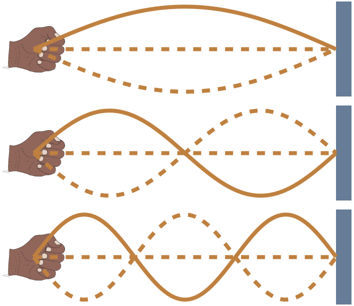
Two traveling waves interfering to make a standing wave.



Two different traveling waves interfering to make a standing wave.

Standing waves: fundamentals and harmonics

- We can characterize a standing wave by the number of nodes (don't move) and anti-nodes (max amplitude).
- For a standing wave on a string, the ends are always nodes.



The first 3 harmonics (fundamental, first overtone, second overtone).

Standing waves and resonance

- We can think of objects as having “resonant frequencies” that correspond to the frequencies (wavelengths) of the fundamental frequency and the harmonics.
- If you make an object vibrate, most of the waves will destructively interfere, but the waves at the right frequencies can setup standing waves that can become quite large.
- You don’t need to shake the end of the object at a particular frequency to create a standing wave inside of it.
- If you pluck a guitar string, you effectively send a bunch of sine waves along the string; those that correspond to the harmonics will constructively interfere and dominate the sound that you hear.

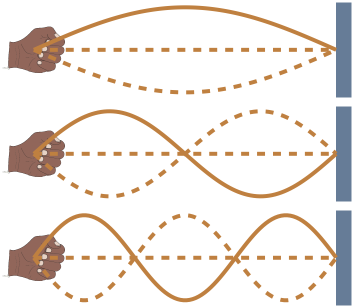
Description of a standing wave

- In contrast to a traveling wave, in a standing wave, each point oscillates with a different amplitude.
- We can describe each point on the string with:

$$D(x,t) = 2A\sin\left(\frac{n\pi}{L}x\right)\cos(\omega t)$$

- Each point oscillates as a simple harmonic oscillator:

$$D(x,t) = A(x)\cos(\omega t)$$



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